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Inventor(s)

CHEN; Feng et al.

RADIO FREQUENCY MODULE, POWER SUPPLY CONTROL METHOD, AND COMMUNICATION DEVICE

Abstract

Provided are a radio frequency module, a power supply control method, and a communication device. The radio frequency module includes at least two power modules, at least two transmission groups and a power supply control circuit. Each transmission group includes at least two power amplifier modules, each power amplifier module is switchably connected to any one of the at least two power modules, and individual power amplifier modules in a same transmission group are configured to perform power amplification on received radio frequency signals under power supply of a same power module. Working statuses of individual power amplifier modules are acquired by the power supply control circuit, and an abnormal transmission group where an abnormal power amplifier module is located may be determined therefrom, and each normal power amplifier module in the abnormal transmission group is controlled to switch to being connected with a target power module.

Inventors: CHEN; Feng (Dongguan, CN), TONG; Lin (Dongguan, CN)

Applicant: GUANGDONG OPPO MOBILE TELECOMMUNICATIONS CORP., LTD.
(Dongguan, CN)

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATION [0001] This application is a continuation of International Application No. PCT/CN2023/122940 filed Sep. 28, 2023, which claims priority to Chinese Patent Application No. 202211519726.7 filed Nov. 30, 2022, all contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of antenna technology, and in particular to a radio frequency module, a power supply control method, and a communication device.

BACKGROUND

[0003] The statements herein merely provide background information related to the present disclosure, and they do not necessarily constitute exemplary techniques.

[0004] With the development of radio frequency technology, a communication device usually includes therein at least two power amplifier modules. The power amplifier (PA) in each power amplifier module is used to amplify a weak signal generated by a radio frequency transceiver into a high-power strong signal, so that the amplified signal is radiated into free space through an antenna for communication with a base station.

[0005] However, in the architecture with at least two power amplifier modules, when an abnormality, such as burning out, occurs in one PA, other PAs powered by a same power module as this one PA may not work properly, which affects the user experience.

SUMMARY

[0006] According to various embodiments of the present disclosure, a radio frequency module, a power supply control method, and a communication device are provided.

[0007] In a first aspect, the embodiments of the present disclosure provide a radio frequency module, including: [0008] at least two power modules; [0009] at least two transmission groups, where each of the transmission groups includes at least two power amplifier modules, each of the power amplifier modules is switchably connected to any one of the at least two power modules, and the individual power amplifier modules in a same transmission group are configured to perform power amplification on received radio frequency signals under power supply of a same power module; and [0010] a power supply control circuit, configured to acquire working statuses of the individual power amplifier modules in each of the transmission groups, determine, based on the working statuses, an abnormal transmission group in which there is an abnormal power amplifier module, and control each normal power amplifier module in the abnormal transmission group to switch from being connected with a currently connected power module to being connected with a target power module, the target power module being any one of the at least two power modules except the currently connected power module.

[0011] In a second aspect, the embodiments of the present disclosure provide a power supply control method, including: [0012] acquiring working statuses of individual power amplifier modules in each transmission group; and [0013] determining, based on the working statuses, an

abnormal transmission group in which there is an abnormal power amplifier module, and controlling each normal power amplifier module in the abnormal transmission group to switch from being connected with a currently connected power module to being connected with a target power module; [0014] where the target power module is any one of at least two power modules except the currently connected power module; there are at least two transmission groups, each of the transmission groups includes at least two power amplifier modules, each of the power amplifier modules is switchably connected to any one of the at least two power modules, and the individual power amplifier modules in a same transmission group are configured to perform power amplification on received radio frequency signals under power supply of a same power module. [0015] In a third aspect, the embodiments of the present disclosure provide a communication device including the radio frequency module as described above. [0016] The details of one or more embodiments of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the present disclosure will be apparent from the description, drawings, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] In order to more clearly illustrate the embodiments of the present disclosure or the technical solutions in the prior art, drawings required for use in the description of the embodiments or the prior art will be briefly introduced below. Apparently, the drawings described below are only some embodiments of the present disclosure. For those of ordinary skill in the art, other drawings can be obtained based on these drawings without paying creative work.

[0018] FIG. 1 is a first structural block diagram of a radio frequency module according to some embodiments.

[0019] FIG. 2 is a second structural block diagram of the radio frequency module according to some embodiments.

[0020] FIG. 3 is a third structural block diagram of the radio frequency module according to some embodiments.

[0021] FIG. 4 is a fourth structural block diagram of the radio frequency module according to some embodiments.

[0022] FIG. 5 is a fifth structural block diagram of the radio frequency module according to some embodiments.

[0023] FIG. 6 is a first flowchart of a power supply control method according to some embodiments.

[0024] FIG. 7 is a second flowchart of the power supply control method according to some embodiments.

[0025] FIG. 8 is a structural block diagram of a communication device according to some embodiments.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0026] In order to make the objects, technical solutions and advantages of the present disclosure more clearly understood, the present disclosure is further described in detail below in conjunction with the drawings and embodiments. It should be understood that the specific embodiments described herein are only used to explain the present disclosure and are not used to limit the present disclosure.

[0027] It can be understood that terms “first”, “second”, etc. used in the present disclosure may be used herein to describe various elements, but these elements are not limited by these terms. These terms are used only to distinguish a first element from another element, and should not be understood as indicating or implying relative importance or implicitly indicating the number of the

involved technical features. Therefore, a feature defined by “first” or “second” may explicitly or implicitly include at least one such feature. In the description of the present disclosure, “multiple/plurality of” means at least two, for example, two, three, etc., unless otherwise clearly and specifically defined.

[0028] It is notable that, when an element is described as being “installed on” another element, it may be directly on the other element or there may be an element located therebetween. When an element is considered to be “connected to/with” another element, it may be directly connected to the another element or there may be also an intermediate element.

[0029] The radio frequency system involved in the embodiments of the present disclosure may be applied to a communication device with a wireless communication function, and such communication device may be a handheld device, a vehicle-mounted device, a smart car, a wearable device, a computing device or other processing devices connected to wireless modems, as well as various user equipment (UE) (for example, mobile phones), mobile stations (MS), etc. For the convenience of description, the above-mentioned devices are collectively referred to as a communication device.

[0030] FIG. 1 is a structural block diagram of a radio frequency module according to some embodiments. Referring to FIG. 1, in the embodiments, the radio frequency module includes at least two power modules **10**, at least two transmission groups **20** and a power supply control circuit **30**.

[0031] Each transmission group **20** includes at least two power amplifier modules **210**, each power amplifier module **210** is switchably connected to any one of the at least two power modules **10**, and the individual power amplifier modules **210** in a same transmission group **20** are configured to perform power amplification on received radio frequency signals under power supply of a same power supply (FIG. 1 only shows a situation in which the power amplifier modules **210** in each transmission group **20** are connected with a power module, and does not show a situation in which the power amplifier modules **210** in each transmission group **20** are disconnected from other power modules **10**). The power supply control circuit **30** is configured to acquire working statuses of the individual power amplifier modules **210** in each transmission group **20**, determine, based on the working statuses, an abnormal transmission group in which there is an abnormal power amplifier module, and control each normal power amplifier module in the abnormal transmission group to switch from being connected with a currently connected power module **10** to being connected with a target power module. The target power module is any one of the at least two power modules **10** except the currently connected power module **10**.

[0032] The number of the power modules **10** is at least two, each power module **10** may be connected with each of the power amplifier modules **210** in a same transmission group **20**, or may also be connected with multiple power amplifier modules **210** in different transmission groups **20**. When the power module **10** is connected with a power amplifier module **210**, the power module **10** provides a power supply signal to the connected power amplifier module **210**, so that the power amplifier module **210** performs power amplification on its received radio frequency signal. In some implementations, the power module **10** may perform voltage regulation according to a frequency band of the radio frequency signal for which the power amplifier module **210** needs to perform power amplification, so as to output a power supply signal suitable for the normal operation of the power amplifier module **210**, thereby meeting the requirements of the user. The power module **10** may include for example a battery and a power management chip (power management IC, PMIC) connected to the battery, where the power of the battery is adjusted and then provided to the power amplifier module **210**.

[0033] The number of the transmission groups **20** is at least two, and each transmission group **20** includes at least two power amplifier modules **210**. As such, since the radio frequency module includes multiple power amplifier modules **210**, the multiple power amplifier modules **210** may provide multiple transmission channels, so that the radio frequency module supports multi-channel

transmission, thereby improving the communication quality and the user experience. In some implementations, the multiple power amplifier modules **210** may be grouped based on whether the radio frequency signals received by the individual power amplifier modules **210** are in a same transmission channel group in actual application, so that multiple power amplifier modules **210** connected to a same transmission channel group may belong to a same transmission group **20**. In some implementations, the multiple power amplifier modules **210** may be grouped based on frequency bands of the radio frequency signals that need power amplification. For example, some power amplifier modules **210** supporting a low-frequency band, a medium-frequency band or a high-frequency band may be in a same transmission group **20**, and some power amplifier modules **210** supporting a high-frequency band or an ultra-high frequency band may be in a same transmission group **20**, so as to support combinations of different frequency bands. The power amplifier modules **210** may also be grouped based on the power supply requirements thereof, for example, different power amplifier modules **210** with similar power supply requirements may be in a same transmission group **20**. It is notable that the embodiments are not limited to the above-mentioned grouping manner, which may be set as required.

[0034] Each power amplifier module **210** is switchably connected to any one of the at least two power modules **10**, and the individual power amplifier modules **210** in a same transmission group **20** are configured to perform power amplification on the received radio frequency signals under the power supply of a same power module. Generally, when all the transmission groups **20** are powered normally, different power modules **10** are connected by default to different transmission groups **20** and thus the power amplifier modules **210** therein. It can be understood that, the switchable connection means that each power amplifier module **210** is capable of being connected to each of the at least two power modules **10** and it is connected to one of the power modules **10** at a time, and the connection between the power amplifier module **210** and each of the power modules **10** may be made available through a switching operation.

[0035] Each power amplifier module **210** may include one or more power amplifiers (PAs). The power terminal of each PA is switchably connected to one of the at least two power modules **10**. The input terminal of each PA is connected to a radio frequency transceiver to receive the radio frequency signal. The output terminal of each PA is connected to an antenna or other radio frequency front-end modules. When a PA is connected with a power module, the PA performs power amplification on a received radio frequency signal and outputs the radio frequency signal after undergoing power amplification to the antenna for transmission outward. In some implementations, the power supply control circuit **30** may include the radio frequency transceiver. When the power supply control circuit **30** includes the radio frequency transceiver, the power supply control circuit **30** may be connected with the input terminal of each power amplifier module **210** to provide a radio frequency signal to each power amplifier module **210** (taking FIG. **1** as an example, it shows the connection between the power supply control circuit **30** and each power amplifier module **210**).

[0036] In some implementations, the power amplifier module **210** may further include other functional components to implement other auxiliary functions, for example, it may further include a low-noise amplifier to further implement a receiving function. When the power amplifier module **210** also includes a low-noise amplifier, the power amplifier module **210** may be a low-frequency power amplifier module with a built-in low-noise amplifier (LB L-PA Mid, Low Band PA Mid With LNA), or a medium-high frequency power amplifier module with a built-in low-noise amplifier (MHB L-PA Mid, Middle and High Band PA Mid With LNA). In addition, it may also be a power amplifier switch module integrated with a filter and a low-noise amplifier (LPAF, LNA-PA ASM module with integrated filter), etc. For example, it may be an LPAF that supports high-frequency signals or ultra-high frequency signals.

[0037] The power supply control circuit **30** is configured to acquire the working statuses of the individual power amplifier modules **210** in each transmission group **20**, determine, based on the

working statuses, an abnormal transmission group in which there is an abnormal power amplifier module, and control each normal power amplifier module in the abnormal transmission group to switch from being connected with a currently connected power module to being connected with a target power module. The target power module is any one of the at least two power modules **10** except the currently connected power module **10**.

[0038] The working status of each power amplifier module **210** includes a normal working status and an abnormal working status. In the power amplifier module **210**, the PA is a high-power device and is often in a high-temperature, high-power and high-current working scenario, and may have a small probability of being burned out. When a power amplifier is burned out, a power line short circuit would occur inside the respective power amplifier module **210**, which may cause other power amplifier modules **210** powered by the same power module **10** to fail to work normally. In the embodiments, the working status of a power amplifier module **210** in which a power line short circuit occurs inside the power amplifier module is defined as an abnormal working status, the power amplifier module **210** in the abnormal working status is defined as an abnormal power amplifier module, and the transmission group **20** to which the abnormal power amplifier module belongs is defined as an abnormal transmission group. It can be understood that other statuses except the abnormal working status may be defined as a normal working status, and the power amplifier module **210** in the normal working status is defined as a normal power amplifier module.

[0039] By acquiring the working statuses of the power amplifier modules **210** through the power supply control circuit **30**, the abnormal power amplifier module(s) and the abnormal transmission group in which the abnormal power amplifier module(s) is(are) located may be determined, and each normal power amplifier module in the abnormal transmission group may be timely controlled to switch from being connected with a currently connected power module **10** to a target power module, thereby ensuring the normal operation of the normal power amplifier module. In this way, the number of available frequency bands of the radio frequency module after the burning out of a power amplifier module **210** is increased, which improves the user experience and reduces customer return rate. It can be understood that, in some other embodiments, some of the multiple transmission groups may include only one power amplifier module. When the abnormal transmission group only includes one abnormal power amplifier module, the power supply control circuit **30** may not perform the related operation of switching to the target power module. In some implementations, the power supply control circuit **30** may obtain the working status of each power amplifier module **210** according to a status parameter of the power amplifier module **210**. The status parameter may include, for example, a voltage parameter, a current parameter or a temperature parameter of the power terminal of the power amplifier module **210**. When a power amplifier module **210** is in the abnormal working status, the power line short circuit would cause a change in the status parameter. For example, the voltage parameter of the power terminal of the power amplifier module **210** may be suddenly changed to zero, and the current parameter and the temperature parameter thereof may increase abnormally. Therefore, the working status of the power amplifier module **210** may be obtained according to the status parameter(s) of the power amplifier module **210**. In some implementations, the power supply control circuit **30** may also acquire the working status of each power amplifier module **210** according to a status signal capable of directly reflecting the status of the power amplifier module **210**. Further, in some implementations, a circuit with corresponding functions may be provided between the power supply control circuit **30** and the power amplifier module **210** to acquire the aforementioned status parameter or status signal. For example, a sampling circuit may be provided, and the sampling circuit samples the status parameter(s) of the power amplifier module **210** and feeds back the status parameter(s) to the power supply control circuit **30**. For example, a detection circuit may be provided, and the detection circuit generates a status signal of the working status according to the status parameter(s) of the power amplifier module **210**, and feeds back the status signal to the power supply control circuit **30**. For example, the power supply control circuit **30** itself may also have a function of

collecting the status parameters. It can be understood that the way of acquiring the working statuses of the power amplifier modules **210** is not limited to those in the above-mentioned embodiments, and embodiments adopting other manners are not further listed here.

[0040] The radio frequency module provided in the embodiments includes at least two power modules **10**, at least two transmission groups **20** and a power supply control circuit **30**. Each transmission group **20** includes at least two power amplifier modules **210**. Each power amplifier module **210** is switchably connected to any one of the at least two power modules **10**. The individual power amplifier modules **210** in a same transmission group **20** are configured to perform power amplification on received radio frequency signals under power supply of a same power module. By acquiring the working status of each power amplifier module **210** through the power supply control circuit **30**, an abnormal transmission group in which there is an abnormal power amplifier module may be determined, and each normal power amplifier module in the abnormal transmission group may be timely controlled to switch from being connected with the currently connected power module **10** to the target power module, thereby ensuring the normal operation of each normal power amplifier module. This increases the number of available frequency bands of the radio frequency module after burning out of a power amplifier module **210**, improves the user experience and reduces the customer return rate.

[0041] In some embodiments, as illustrated in FIG. 2, the radio frequency module further includes a switch module **40**, and the power supply control circuit **30** controls each power amplifier module **210** to be connected with or disconnected from a power module **10** by controlling the on/off state of the switch module **40**.

[0042] The switch module **40** has a controlled terminal, multiple first terminals and multiple second terminals. Each first terminal of the switch module **40** is connected to one power module **10**, and at least some of the multiple first terminals are connected to a same power module **10**. Each second terminal of the switch module **40** is connected to the power terminal of one power amplifier module **210**. The controlled terminal of the switch module **40** is connected to the power supply control circuit **30**. The switch module **40** is configured to switch, under control of the power supply control circuit **30**, the normal power amplifier module in the abnormal transmission group from being connected with the currently connected power module **10** to being connected with the target power module.

[0043] The switch module **40** is connected to the power supply control circuit **30** through the controlled terminal, connected to the multiple power modules **10** respectively through the multiple first terminals, and connected to the multiple power amplifier modules **210** respectively through the multiple second terminals. Under the control of the power supply control circuit **30**, the switch module **40** switches the normal power amplifier module in the abnormal transmission group to being connected with the target power module.

[0044] In some implementations, a default power module **10** may be selected as the currently connected power module. When each power amplifier module **210** is in the normal working status, each power amplifier module **210** is brought into connection with its default power module **10** through a switching operation of the switch module **40**, so that each default power module **10** supplies power to its default connected transmission group **20**. When there is an abnormal power amplifier module, the switch module **40** performs a switching operation to disconnect other normal power amplifier modules in the transmission group **20** where the abnormal power amplifier module is located from the default power module **10**, and also bring the normal power amplifier modules to being connected with other power modules **10**.

[0045] By means of the power supply control circuit **30** and the switch module **40**, when an abnormality occurs in a power amplifier module **210**, each normal power amplifier modules in the abnormal transmission group can be switched to being connected with the target power module in time, thereby ensuring that such normal power amplifier module can continue working normally.

[0046] In some embodiments, the switch module **40** is further configured to disconnect the

abnormal power amplifier module from its currently connected power module, under the control of the power supply control circuit **30**.

[0047] When a PA inside a power amplifier module **210** is burned out, an abnormally large current would be generated in the power supply path between the abnormal power amplifier module and its connected power module. When the power supply control circuit **30** determines the abnormal power amplifier module based on the working statuses of the individual power amplifier modules **210**, the switch module **40** is timely controlled to disconnect the abnormal power amplifier module from its currently connected power module, so as to avoid an abnormally large current from being continuously generated in the power amplifier module **210** which would otherwise lead to severe heating.

[0048] In some embodiments, as illustrated in FIG. 2, the power supply control circuit **30** may include a radio frequency transceiver **310** and a controller **320**.

[0049] The controller **320** is configured to acquire the working status of each power amplifier module **210**, and generate a power switching instruction when the working status is an abnormal working status, the power switching instruction carrying parameter information of the abnormal transmission group and the normal power amplifier module(s) therein. The radio frequency transceiver **310** is connected with each of the controller **320**, the input terminal of each power amplifier module **210**, and the controlled terminal of the switch module **40**. The radio frequency transceiver **310** is configured to provide the radio frequency signals, and control, according to the power switching instruction, the switch module **40** to switch each normal power amplifier module in the abnormal transmission group from being connected with the currently connected power module **10** to being connected with the target power module.

[0050] The controller **320** is connected with the radio frequency transceiver **310**. The controller **320** can acquire the working status of each power amplifier module **210**, and generate a power switching instruction when the working status is an abnormal working status, to instruct the radio frequency transceiver **310** to perform a corresponding operation. The manner in which the controller **320** acquires the working status of each power amplifier module **210** may refer to the relevant description in the above embodiments, which will not be repeated here. In some implementations, when the radio frequency module is applied in a communication device, the controller **320** may be an application processor (AP) or a baseband processor in the communication device. Taking the controller **320** as an AP as an example, when the AP finds an abnormal power amplifier module based on the working statuses of the individual power amplifier modules **210**, it notifies, by means of the power switching instruction, the radio frequency transceiver **310** to perform a corresponding operation.

[0051] The radio frequency transceiver **310** is connected with each of the controller **320**, the input terminal of each power amplifier module **210**, and the controlled terminal of the switch module **40**. The radio frequency transceiver **310** may provide radio frequency signals to the power amplifier modules **210**, and may also receive the power switching instruction output by the controller **320**. Since the power switching instruction carries the parameter information of the abnormal transmission group and the normal power amplifier module(s), the radio frequency transceiver **310** may generate, based on the parameter information, a corresponding control signal to control the switch module **40** to performing the switching to the target power module.

[0052] Through the interaction between the controller **320** and the radio frequency transceiver **310**, the abnormal power amplifier module may be found in time, and the on/off state of the switch module **40** may be controlled in time, so that the switch module **40** may switch in time the normal power amplifier module(s) in the abnormal transmission group to being connected with the target power module, thereby ensuring the normal operation of the normal power amplifier module(s).

[0053] In some embodiments, still referring to FIG. 2, the switch module **40** includes multiple switch units **410**, and the radio frequency transceiver **310** controls the on/off state of each switch unit **410** to switch the normal power amplifier module(s) in the abnormal transmission group to

being connected with the target power module.

[0054] Each of the multiple switch units **410** has a controlled terminal, at least two first terminals and a second terminal (two first terminals and one second terminal are illustrated as an example in FIG. 2). The at least two first terminals of each switch unit **410** are respectively connected to the at least two power modules **10**, and the second terminal of each switch unit **410** is connected to one power amplifier module **210**. The radio frequency transceiver **310** is configured with multiple control interfaces, and each control interface is connected to the controlled terminal of one switch unit **410**. The radio frequency transceiver **310** is configured to generate a control signal according to the power switching instruction, and send the control signal to a target switch unit **410** through a respective control interface, so as to control the target switch unit **410** to switch the normal power amplifier module in the abnormal transmission group from being connected with its currently connected power module to being connected with the target power module, where the target switch unit **410** is connected with the normal power amplifier module.

[0055] The switch unit **410** is provided on the connection path between each power amplifier module **210** and the power modules **10**. By controlling the on/off states of the switch units **410**, the power supply of the power modules **10** to each power amplifier module **210** may be controlled. The control signal includes a turn-on signal and a turn-off signal. The turn-on signal is used to control the switch unit **410** to bring the respective connection path that was originally in a disconnected state into being in a connected state, and the turn-off signal is used to control the switch unit **410** to bring the respective connection path that was originally in a connected state into being in a disconnected state.

[0056] When the radio frequency transceiver **310** receives a power switching instruction, the radio frequency transceiver **310** determines, based on the parameter information carried in the power switching instruction, a target switch unit **410** that needs to perform the switching to the target power module, generates a control signal and sends the control signal through a respective control interface to the corresponding target switch unit **410**, so as to control the target switch unit **410** to perform the switching to the target power module. By providing the multiple switch units **410** with each switch unit **410** being arranged on the connection path between each power amplifier module **210** and the power modules **10**, the on/off state of each switch unit **410** may be independently controlled, and the switching to the target power module can be performed more accurately.

[0057] In some implementations, the control interface of the radio frequency transceiver **310** includes a mobile industry processor interface (MIPI) or a general purpose input output (GPIO) interface. The radio frequency transceiver **310** may send a control signal to the switch unit **410** by controlling the level of the MIPI or GPIO interface to be changed, thereby improving the control efficiency.

[0058] In some implementations, each switch unit **410** includes at least one switch element, which may be for example a single-pole multi-throw switch. For example, when the power amplifier module **210** is capable of being connected with two power modules **10**, the switch unit **410** may be a single-pole double-throw switch, and the two first terminals of the single-pole double-throw switch are respectively connected to the output terminals of the two power modules **10**, and the second terminal of the single-pole double-throw switch is connected to the power terminal of the power amplifier module **210**. The single-pole double-throw switch may be configured to make one of the power modules **10** connected with the power amplifier module **210** by default, and make the other power module **10** disconnected from the power amplifier module **210**. When the power supply control circuit **30** determines an abnormal power amplifier module, the single-pole double-throw switch disconnects the normal power amplifier module from the power module **10** that also supplies power to the abnormal power amplifier module, and connects the normal power amplifier module to the other power module **10**, thereby achieving the switching to the target power module.

[0059] In some embodiments, the switch unit **410** may be provided outside the power amplifier module **210** (in the embodiment of FIG. 2, it is illustrated by taking a case where the switch unit

410 is external to the power amplifier module as an example, and auxiliary reference may be made to FIG. 2). Each power amplifier module **210** may be configured with a power interface and an input interface, the power interface is connected to the second terminal of the switch unit **410**, and the input interface is connected to a transmission interface of the radio frequency transceiver **310**. The power interface is used to receive the power supply signal of the power module **10**, and the input interface is used to receive the radio frequency signal output by the radio frequency transceiver **310**.

[0060] In some embodiments, the switch unit **410** may be integrated in the power amplifier module **210**. As illustrated in FIG. 3, each power amplifier module **210** includes a power amplifier, and each power amplifier module **210** is configured with at least two power interfaces, a controlled interface and an input interface (in FIG. 3, it is illustrated by taking two power interfaces as an example, the two power interfaces are VCC1 and VCC2, the input interface is PA IN, and the controlled interface is BC; the power amplifier and the switch unit **410** are not shown in the figure). The at least two power interfaces are respectively connected to the at least two power modules **10**, the controlled interface is connected to a control interface of the radio frequency transceiver **310**, and the input interface is connected with each of a transmission interface of the radio frequency transceiver **310** and an input terminal of the power amplifier. Each switch unit **410** is integrated in a corresponding power amplifier module **210**, the at least two first terminals of the switch unit **410** are respectively connected to the at least two power interfaces, the second terminal of the switch unit **410** is connected to a power terminal of the power amplifier, and the controlled terminal of the switch unit **410** is connected to the controlled interface.

[0061] The power amplifier module **210** is configured with the at least two power interfaces, the controlled interface and the input interface. Each switch unit **410** is integrated in the corresponding power amplifier module **210**, the first terminals of the switch unit **410** are connected to the external power modules **10** through the power interfaces, the second terminal of the switch unit **410** is connected to the power amplifier of the power amplifier module **210**, and the controlled terminal of the switch unit **410** is connected to the external radio frequency transceiver **310** through the controlled interface. By integrating the switch unit **410** inside the power amplifier module **210**, the area of the mainboard occupied by the radio frequency module can be reduced, the integration level can be improved, and it is beneficial to the miniaturization of the radio frequency module and the reduction of costs.

[0062] It can be understood that, when the power amplifier module **210** includes multiple power amplifiers, the switch unit **410** may also have multiple second terminals, and each second terminal is connected to a power terminal of one power amplifier. It can be understood that the power amplifier module **210** is further configured with an output interface, and the output interface may be connected to an antenna through a radio frequency front-end circuit.

[0063] In some other embodiments, the switch unit **410** may also be integrated into the power module **10**. For example, multiple switch units **410** corresponding to the default power module **10** in the above embodiments may be integrated inside the default power module **10** to improve the integration level, which is also conducive to the miniaturization of the radio frequency module and the reduction of costs.

[0064] In some embodiments, the radio frequency module may further include a status detection circuit (for example, as indicated by **50** in FIG. 2), and the power supply control circuit **30** acquires the working status of each power amplifier module **210** through the status detection circuit.

[0065] The status detection circuit is connected to each of the controller **320** and the power amplifier module **210**. The status detection circuit is configured to detect the working status of each power amplifier module **210** and feedback the working status to the controller **320**. Specifically, the status detection circuit may detect the status parameter(s) of the power amplifier module **210**, determine the working status of the power amplifier module **210** based on the status parameter(s), and feedback the working status to the controller **320** for acquisition by the controller **320**.

[0066] In some implementations, the status detection circuit may generate, based on the status detection result, a signal indicating a change of a status flag bit. For example, when an abnormal power amplifier module is detected, a high or low level signal is generated. The high or low level signal is used to indicate to the controller **320** that a corresponding power amplifier module **210** is in an abnormal working status. In some implementations, the status detection circuit may be a voltage detection circuit, a current sampling circuit, a temperature detection circuit, etc., which is used to detect the working status of the power amplifier module **210** based on the status parameter(s) of the power amplifier module **210**. It can be understood that, when the status detection circuit is a voltage detection circuit or a current sampling circuit, the status detection circuit may be connected with the power amplifier module **210**; and when the status detection circuit is a temperature detection circuit, the status detection circuit may not be connected with the power amplifier module **210**, and the temperature parameter may be obtained by detecting the temperature of the environment surrounding the power amplifier module **210**.

[0067] In some implementations, there may be one status detection circuit, and the status detection circuit may have multiple detection terminals and multiple feedback terminals. Each detection terminal may be connected to one power amplifier module **210**, and each feedback terminal is connected to one feedback receiving terminal of the controller **320**. The status detection circuit is configured to feedback the working status obtained by each detection terminal to the corresponding feedback terminal, so that the controller **320** may determine the working status of each power amplifier module **210** based on the status signals received by the feedback receiving terminals.

[0068] In some implementations, there may be multiple status detection circuits. As illustrated in FIG. 4, each power amplifier module **210** is further configured with a status feedback interface (as indicated by BD in the figure). Each status detection circuit is integrated in one corresponding power amplifier module **210**, and the feedback terminal of each status detection circuit is connected to a detection interface of the controller **320** through the status feedback interface, and the detection terminal of each status detection circuit is connected to the power terminal of the power amplifier (the power amplifier and the status detection circuit are not shown in the figure).

[0069] By integrating each status detection circuit inside the power amplifier module **210**, in one aspect, the area of the mainboard occupied by the radio frequency module can be reduced, the integration level can be improved, and it is beneficial to the miniaturization of the radio frequency module and the reduction of costs. In another aspect, a detection distance between the status detection circuit and the power terminal of the power amplifier can be shortened, thereby improving the detection accuracy of the status detection circuit.

[0070] In some embodiments, the power supply control circuit **30** is further configured to acquire a target frequency band of the radio frequency signal for which the power amplification is supported by the abnormal power amplifier module, and update an available network search frequency band based on the target frequency band. The available network search frequency band is a frequency band at which the radio frequency module supports for radio frequency signal transmission in a target standard network.

[0071] The target frequency band is a frequency band of a radio frequency signal for which the abnormal power amplifier module supports power amplification. When a power amplifier module **210** is in an abnormal status, it means that the target frequency band supported by the power amplifier module **210** is no longer available.

[0072] The available network search frequency band is a frequency band at which the radio frequency module supports radio frequency signal transmission in the target standard network. The updating the available network search frequency band based on the target frequency band may include removing the target frequency band from the original available network search frequency band. The target standard network may be in one or more network standards, including for example Global System for Mobile Communications (GSM), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE) and New Radio (NR), and each network standard may

include multiple frequency bands, which is not further limited in the embodiments. Taking the LTE network standard and NR network standard as an example, each of an LTE signal and a NR signal includes a low-frequency signal, a medium-frequency signals, a high-frequency signal and an ultra-high frequency signal. The frequency band division of the low-frequency signal, the medium-frequency signal, the high-frequency signal and the ultra-high frequency signal may be referred to the following table.

TABLE-US-00001

LTE signal	NR signal	Frequency Band	frequency band	frequency band	L: Low					
frequency B5, B8, B12, B13, N5, N8, N12, N13, B14, B17, B18, B19, N14, N17, N18, N19, B20, B26, B28A	N20, N26, N28A	M: Medium frequency	B1, B2, B3, B4, N1, N2, N3, N4, B25, B34, B39, B66	N25, N34, N39, N66	H: High frequency	B7, B30, B38, N7, N30, N38, B40, B41	N40, N41	UH: Ultra-high	N77, N78, N79	frequency

[0073] It is notable that the 5G network uses the same frequency band as 4G, and only the identifier before the number is changed. In addition, the 5G network has added some ultra-high frequency bands that are not available in the 4G network, such as N77, N78 and N79.

[0074] When the radio frequency module is applied in a communication device, the communication device usually needs to search for a network during operation. In the related art, the network search process usually attempts to search all frequency bands supported by the radio frequency module.

When a power amplifier module **210** is in an abnormal status, it means that the target frequency band supported by the power amplifier module **210** is no longer available, and the searching for the unavailable frequency band may cause a crashing risk of the device. In the embodiments, the updated available network search frequency band removes the target frequency band, which allows the device to skip the unavailable frequency band when performing a network search operation. Thus, this not only reduces the power consumption caused by the network search, but also shortens the network search time and speeds up the communication service.

[0075] In some implementations, the controller **320** in the power supply control circuit **30** may acquire the target frequency band of the radio frequency signal for which power amplification is supported by the abnormal power amplifier module, and update the available network search frequency band based on the target frequency band. In some implementations, the controller **320** may also establish a network search list based on the updated available network search frequency band, so that the device may perform the network search operation according to the network search list. By establishing the network search list, the network search speed of the device can be further improved.

[0076] The above embodiments are further described below by taking, as an example, a case where the at least two power modules include a first power module and a second power module, and the at least two transmission groups include a first transmission group and a second transmission group.

[0077] The first transmission group includes a low-frequency power amplifier module, a medium-high frequency amplifier module and a first ultra-high frequency amplifier module each capable of being connected to the first power module and the second power module. The low-frequency power amplifier module is configured to support power amplification of low-frequency signals, the medium-high frequency amplifier module is configured to support power amplification of medium-frequency signals and high-frequency signals, and the first ultra-high frequency amplifier module is configured to support power amplification of ultra-high frequency signals. Each power amplifier module in the first transmission group is connected to the first power module by default. The second transmission group includes a high-frequency amplifier module and a second ultra-high frequency amplifier module each capable of being connected to the first power module and the second power module. The high-frequency amplifier module is configured to support power amplification of high-frequency signals, and the second ultra-high frequency amplifier module is configured to support power amplification of ultra-high frequency signals. Each power amplifier module in the second transmission group is connected to the second power module by default.

[0078] The power supply control circuit **30** is configured to: when the first transmission group includes at least one abnormal power amplifier module and the second transmission group does not include an abnormal power amplifier module, switch each normal power amplifier module in the first transmission group from being connected with the currently connected first power module to being connected with the second power module; alternatively, when the second transmission group includes at least one abnormal power amplifier module and the first transmission group does not include an abnormal power amplifier module, switch each normal power amplifier module in the second transmission group from being connected with the currently connected second power module to being connected with the first power module.

[0079] As illustrated in FIG. 5 (in this figure, it is illustrated by taking, as an example, a case where the power supply control circuit **30** includes the radio frequency transceiver **310** and the controller **320**, the radio frequency module further includes the status detection circuit and the switch module **40**, and the switch module **40** includes multiple switch units **410**), the low-frequency power amplifier module, the medium-high frequency amplifier module and the first ultra-high frequency amplifier module of the first transmission group are powered by the first power module by default, and the high frequency amplifier module and the second ultra-high-frequency amplifier module of the second transmission group are powered by the second power module by default. The first power module is power supply **1**, and the second power module **10** is power supply **2**. The low-frequency amplifier module may be an LB LPAMID module, the medium-high frequency amplifier module may be an MHB LPAMID module, and the first ultra-high frequency amplifier module may be an N78 LPAF #1 module. The high-frequency amplifier module may be an N41 LPAF PA module, and the second ultra-high frequency amplifier module may be an N78 LAPF #2 module. The controller **320** may be an AP. The LB LPAMID module, the MHB LPAMID module, the N78 LPAF #1 module, the N41 LPAF PA module and the N78 LAPF #2 module each may be connected to a corresponding antenna (ANT1, ANT2, ANT3, ANT4 and ANT5 in the figure) through a front-end module.

[0080] Assuming that a short circuit occurs in the LB LPAMID module due to burning out of PA, the AP determines, by means of the status detection circuit, that the LB LPAMID module is in an abnormal working status, determines the first transmission group as an abnormal transmission group, and then generates a power switching instruction to notify the radio frequency transceiver **310**. The radio frequency transceiver **310** generates a control signal according to the power switching instruction, and sends, through the MIPI or GPIO interface, the control signal to the switch units **410** corresponding to the MHB LPAMID module and the N78 LPAF #1 module, to control the switch units **410** to switch the currently connected first power module to the second power module. In addition, the AP may remove low-frequency bands, such as B5, B8, B28, N5, N8, and N28, from the network search list.

[0081] Assuming that a short circuit occurs in the N78 LAPF #2 module due to burning out of PA, the AP determines, by means of the status detection circuit, that the N78 LAPF #2 module is in an abnormal working status, determines the second transmission group as an abnormal transmission group, and then generates a power switching instruction to notify the radio frequency transceiver **310**. The radio frequency transceiver **310** generates a control signal according to the power switching instruction, and sends, through the MIPI or GPIO interface, the control signal to the switch unit **410** corresponding to the N41 LPAF module, to control the switch unit **410** to switch the currently connected second power module to the first power module. In addition, the AP may remove the N78 ultra-high frequency band from the network search list.

[0082] It can be understood that the probability of short circuit caused due to burning out of PA is usually low, and there is also a low probability that the first emission group and the second emission group simultaneously have an abnormal power amplifier module therein. Therefore, the target power module is usually switched between the first power module and the second power module. In order to avoid the extremely small probability that the first emission group and the

second emission group simultaneously have an abnormal power amplifier module therein, a third power module may be added to serve as an alternative for the switching to the target power module.

[0083] The division of various modules and circuits in the above-mentioned radio frequency module is only for illustration. In other embodiments, the radio frequency module may be divided into different circuits as needed to implement all or part of the functions of the above-mentioned radio frequency module.

[0084] FIG. 6 is a method flow chart of a power supply control method according to some embodiments. As illustrated in FIG. 6, in the embodiments, the power supply control method includes operations **602** to **604**.

[0085] At **602**, working statuses of individual power amplifier modules in each transmission group are acquired.

[0086] At **604**, an abnormal transmission group in which there is an abnormal power amplifier module is determined based on the working statuses, and each normal power amplifier module in the abnormal transmission group is controlled to switch from being connected with a currently connected power module to being connected with a target power module.

[0087] The target power module is any one of at least two power modules except the currently connected power module. There are at least two transmission groups, each transmission group includes at least two power amplifier modules, and each power amplifier module is switchably connected to any one of the at least two power modules. The individual power amplifier modules in a same transmission group are configured to perform power amplification on received radio frequency signals under the power supply of a same power module.

[0088] For the transmission groups, the power amplifier modules, the power modules, and the target power module, etc., reference may be made to the relevant description in the above embodiments, which will not be repeated here. Operations **602** to **604** may be executed by the power supply control circuit in the above embodiments, and the details may be referred to the relevant description in the above embodiments, which will not be repeated here.

[0089] In some implementations, the operation of controlling each normal power amplifier module in the abnormal emission group to switch to being connected with the target power module may include: controlling a switch module to switch each normal power amplifier module in the abnormal emission group from being connected with the currently connected power module to the target power module. In some implementations, the working status of each power amplifier module may be obtained based on feedback from a status detection circuit. For the switch module and the status detection circuit, reference may be made to the relevant description in the above embodiments, which will not be repeated here.

[0090] In the power supply control method provided in the embodiments, an abnormal transmission group in which there is an abnormal power amplifier module may be determined by acquiring the working status of each power amplifier module, so as to timely control each normal power amplifier module in the abnormal transmission group to switch to being connected with the target power module, thereby ensuring that the normal power amplifier module can continue working normally. In this way, the number of available frequency bands of the radio frequency module after the burning out of a power amplifier module is increased, which improves the user experience and reduces the customer return rate.

[0091] In some embodiments, as illustrated in FIG. 7, the power supply control method further includes operation **702**.

[0092] At **702**, the abnormal power amplifier module is disconnected from the currently connected power module.

[0093] This operation may be executed by the power supply control circuit and the switch module in the above embodiments. The details thereof may refer to the relevant description of the above embodiment, which will not be repeated here.

[0094] In some embodiments, as illustrated in FIG. 7, the power supply control method further includes operation **704**.

[0095] At **704**, a target frequency band of a radio frequency signal for which power amplification is supported by the abnormal power amplifier module is acquired, and an available network search frequency band is updated based on the target frequency band. The available network search frequency band is a frequency band at which the radio frequency module supports radio frequency signal transmission in a target standard network.

[0096] Operation **704** may be executed by the power supply control circuit in the above embodiment. The details thereof may refer to the relevant description of the above embodiments, which will not be repeated here.

[0097] It is understandable that, the operations in the flowchart of the above embodiments are displayed in sequence as indicated by the arrows, but these operations are not necessarily executed in the order indicated by the arrows. Unless otherwise specified herein, there is no strict order limitation for the execution of these operations, and these operations may be executed in other orders. Moreover, at least part of the operations in the flowchart of the above embodiments may include multiple sub-operations or multiple stages. These sub-operations or stages are not necessarily executed at the same time, but may be executed at different times. The execution order of these sub-operations or stages is not necessarily sequential, but may be executed in turn or alternately with other operations or at least part of the sub-operations or stages of other operations.

[0098] The embodiments of the present disclosure further provide a communication device, which may include the radio frequency module in any of the above embodiments. The communication device can timely control each normal power amplifier module in the abnormal transmission group to switch to being connected with the target power module, thereby ensuring the normal operation of the normal power amplifier module. This increases the number of available frequency bands of the radio frequency module after the burning out of a power amplifier module, which improves the user experience and reduces the customer return rate.

[0099] The embodiments of the present disclosure further provide a communication device, which may include at least two power modules, at least two transmission groups, a memory, and a processor. For the power modules and the transmission groups, reference may be made to the relevant description in the above embodiments, which will not be repeated here. The memory stores a computer program which, when being implemented by the processor causes the operations of the power supply control method in the above embodiment to be implemented.

[0100] The communication device of the embodiments includes at least two power modules, at least two transmission groups, a memory and a processor. The communication device can timely control each normal power amplifier module in the abnormal transmission group to switch to being connected with the target power module, thereby ensuring the normal operation of the normal power amplifier module. This increases the number of available frequency bands of the radio frequency module after the burning out of a power amplifier module, which improves the user experience and reduces the customer return rate.

[0101] As illustrated in FIG. 8, the above-mentioned communication device is further illustrated by taking it as a mobile phone **11** as an example. Specifically, as illustrated in FIG. 8, the mobile phone **11** may include a memory **21** (which optionally includes one or more computer-readable storage medium), a processor **22**, a peripheral interface **23**, a radio frequency system **24**, and an input/output (I/O) subsystem **26**. These components optionally communicate with each other via one or more communication buses or signal lines **29**. Those skilled in the art would appreciate that the mobile phone **11** shown in FIG. 8 does not constitute a limitation on the mobile phone, and may include more or fewer components than those shown in the figure, or a combination of certain components, or a different arrangement of the components. The various components shown in FIG. 8 are implemented in hardware, software, or a combination of hardware and software, including one or more signal processing and/or application-specific integrated circuits.

[0102] The memory **21** optionally includes a high-speed random access memory, and also optionally includes a non-volatile memory, such as one or more disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Exemplarily, the software components stored in the memory **21** include an operating system **211**, a communication module (or an instruction set) **212**, a global positioning system (GPS) module (or an instruction set) **213**, and the like.

[0103] The processor **22** and other control circuits (such as a control circuit in the radio frequency system **24**) may be used to control the operations of the mobile phone **11**. The processor **22** may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors, power management units, audio codec chips, application-specific integrated circuits, and the like.

[0104] The processor **22** may be configured to implement a control algorithm that controls the use of the antenna in the mobile phone **11**. The processor **22** may also issue control commands for controlling switches in the radio frequency system **24**, etc.

[0105] The I/O subsystem **26** couples input/output peripherals on the mobile phone **11**, such as a keypad and other input control devices, to the peripheral interface **23**. The I/O subsystem **26** optionally includes a touch screen, keys, a tone generator, an accelerometer (a motion sensor), an ambient light sensor and other sensors, a light emitting diode and other status indicators, data interfaces, and the like. Exemplarily, a user may control the operations of the mobile phone **11** by providing commands through the I/O subsystem **26**, and may receive status information and other output from the mobile phone **11** by using the output resources of the I/O subsystem **26**. For example, the user may press a button **261** to turn on or off the mobile phone.

[0106] The radio frequency system **24** may include the radio frequency module in any of the aforementioned embodiments.

[0107] The embodiments of the present disclosure further provide a computer-readable storage medium, particularly one or more non-transitory computer-readable storage media containing computer-executable instructions which, when being executed by one or more processors, cause the processors to perform the operations of the power supply control method.

[0108] The embodiments of the present disclosure further provide a computer program product including instructions which, when being executed on a computer, causes the computer to implement the operations of the power supply control method.

[0109] Any memory, storage, database, or other medium referred to herein may include a non-volatile and/or volatile memory. A suitable non-volatile memory may include read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), or a flash memory. The volatile memory may include random access memory (RAM), which acts as an external cache memory. By way of illustration and not limitation, RM is available in a variety of forms, such as static RM (SRM), dynamic RM (DRM), synchronous DRM (SDRM), double data rate SDRM (DDR SDRM), enhanced SDRM (ESDRM), synchronous link (Synchlink) DRM (SLDRM), memory bus (Rmbus) direct RM (RDRM), direct memory bus dynamic RM (DRDRM), and memory bus dynamic RM (RDRM).

[0110] The technical features of the above embodiments may be combined arbitrarily. To make the description concise, not all possible combinations of the technical features in the above embodiments are described. However, as long as there is no contradiction in the combination of these technical features, they should be considered to be within the scope of the present disclosure.

[0111] The above embodiments only illustrate several implementations of the present disclosure, and the descriptions thereof are relatively specific and detailed, but they should not be construed as limiting the scope of protection of the present disclosure. It should be noted that, for those of ordinary skill in the art, several modifications and improvements can be made without departing from the concept of the present disclosure, which all fall within the scope of protection of the

present disclosure. Therefore, the scope of protection of the disclosure should be subjected to the attached claims.

Claims

1. A radio frequency module, comprising: at least two power modules; at least two transmission groups, wherein each of the transmission groups comprises at least two power amplifier modules, each of the power amplifier modules is switchably connected to any one of the at least two power modules, and the individual power amplifier modules in a same transmission group are configured to perform power amplification on received radio frequency signals under power supply of a same power module; and a power supply control circuit, wherein the power supply control circuit is configured to acquire working statuses of the individual power amplifier modules in each of the transmission groups, determine, based on the working statuses, an abnormal transmission group in which there is an abnormal power amplifier module, and control each normal power amplifier module in the abnormal transmission group to switch from being connected with a currently connected power module to being connected with a target power module, the target power module being any one of the at least two power modules except the currently connected power module.
2. The radio frequency module as claimed in claim 1, further comprising: a switch module having a controlled terminal, a plurality of first terminals and a plurality of second terminals, wherein each of the plurality of first terminals of the switch module is connected to one of the power modules, at least some of the plurality of first terminals are connected to a same power module, each of the plurality of second terminals of the switch module is connected to a power terminal of one of the power amplifier modules, the controlled terminal of the switch module is connected to the power supply control circuit, and the switch module is configured to switch, under control of the power supply control circuit, the normal power amplifier module in the abnormal transmission group from being connected with the currently connected power module to the target power module.
3. The radio frequency module as claimed in claim 2, wherein the switch module is further configured to disconnect, under the control of the power supply control circuit, the abnormal power amplifier module from the currently connected power module.
4. The radio frequency module as claimed in claim 2, wherein the power supply control circuit comprises: a controller, wherein the controller is configured to acquire the working status of each of the power amplifier modules, and generate a power switching instruction in response to the working status being an abnormal working status, the power switching instruction carrying parameter information of the abnormal transmission group and the normal power amplifier module in the abnormal transmission group; and a radio frequency transceiver, wherein the radio frequency transceiver is connected with each of the controller, input terminals of the power amplifier modules, and the controlled terminal of the switch module, and the radio frequency transceiver is configured to provide the radio frequency signals, and control, according to the power switching instruction, the switch module to switch the normal power amplifier module in the abnormal transmission group from being connected with the currently connected power module to being connected with the target power module.
5. The radio frequency module as claimed in claim 4, wherein the switch module comprises: a plurality of switch units, wherein each of the plurality of switch units has a controlled terminal, at least two first terminals and a second terminal, the at least two first terminals of each of the plurality of switch units are respectively connected to the at least two power modules, and the second terminal of each of the plurality of switch units is connected to one of the power amplifier modules; wherein the radio frequency transceiver is configured with a plurality of control interfaces, each of the plurality of control interfaces is connected with the controlled terminal of one of the plurality of switch units; the radio frequency transceiver is configured to generate a control signal according to the power switching instruction, and send the control signal to a target

switch unit through a respective control interface, to control the target switch unit to switch the normal power amplifier module in the abnormal transmission group from being connected with the currently connected power module to being connected with the target power module, the target switch unit being connected with the normal power amplifier module in the abnormal transmission group.

6. The radio frequency module as claimed in claim 5, wherein each of the power amplifier modules comprises a power amplifier, and each of the power amplifier modules is configured with at least two power interfaces, a controlled interface and an input interface, the at least two power interfaces are respectively connected to the at least two power modules, the controlled interface is connected with a control interface of the radio frequency transceiver, and the input interface is connected with each of a transmission interface of the radio frequency transceiver and an input terminal of the power amplifier; wherein each of the plurality of switch units is integrated in a corresponding power amplifier module, the at least two first terminals of each switch unit are respectively connected with the at least two power interfaces of the corresponding power amplifier module, the second terminal of each switch unit is connected with a power terminal of the power amplifier of the corresponding power amplifier module, and the controlled terminal of each switch unit is connected with the controlled interface of the corresponding power amplifier module.

7. The radio frequency module as claimed in claim 5, wherein each of the power amplifier modules is configured with a power interface and an input interface, the power interface is connected with the second terminal of a corresponding switch unit, and the input interface is connected with a transmission interface of the radio frequency transceiver; the power interface is configured to receive a power supply signal of a power module connected with the power amplifier module, and the input interface is configured to receive the radio frequency signal output by the radio frequency transceiver.

8. The radio frequency module as claimed in claim 4, further comprising: at least one status detection circuit, wherein the at least one status detection circuit is connected with each of the controller and the power amplifier modules, and is configured to detect the working statuses of the power amplifier modules and feedback the working statuses to the controller.

9. The radio frequency module as claimed in claim 8, wherein each of the power amplifier modules is further configured with a status feedback interface, and each of the power amplifier modules comprises a power amplifier; the at least one status detection circuit comprises a plurality of status detection circuits, each of the plurality of status detection circuits is integrated in a corresponding power amplifier module, a feedback terminal of each of the status detection circuits is connected with a detection interface of the controller through the status feedback interface of the corresponding power amplifier module, and a detection terminal of each of the status detection circuits is connected with a power terminal of the power amplifier of the corresponding power amplifier module.

10. The radio frequency module as claimed in claim 8, wherein the at least one status detection circuit comprises one status detection circuit, the status detection circuit has a plurality of detection terminals and a plurality of feedback terminals, each of the plurality of detection terminals is connected with one of the power amplifier modules, and each of the plurality of feedback terminals is connected with one of feedback receiving terminals of the controller; the status detection circuit is configured to feedback, from a corresponding feedback terminal, the working status obtained through detection of each of the detection terminals; and the controller is configured to determine the working statuses of the individual power amplifier modules, based on status signals received by the feedback receiving terminals.

11. The radio frequency module as claimed in claim 1, wherein the power supply control circuit is further configured to acquire a target frequency band of a radio frequency signal for which power amplification is supported by the abnormal power amplifier module, and update an available network search frequency band based on the target frequency band, the available network search

frequency band being a frequency band at which the radio frequency module supports for radio frequency signal transmission in a target standard network.

12. The radio frequency module as claimed in claim 1, wherein the at least two power modules comprise a first power module and a second power module, and the at least two transmission groups comprise: a first transmission group, comprising a low-frequency power amplifier module, a medium-high frequency amplifier module and a first ultra-high frequency amplifier module each capable of being connected with the first power module and the second power module, wherein the low-frequency power amplifier module is configured to support power amplification of low-frequency signals, the medium-high frequency amplifier module is configured to support power amplification of medium-frequency signals and high-frequency signals, the first ultra-high frequency amplifier module is configured to support power amplification of ultra-high frequency signals, and each power amplifier module in the first transmission group is connected with the first power module by default; and a second transmission group, comprising a high-frequency amplifier module and a second ultra-high frequency amplifier module each capable of being connected with the first power module and the second power module, wherein the high-frequency amplifier module is configured to support power amplification of the high-frequency signals, the second ultra-high frequency amplifier module is configured to support power amplification of the ultra-high frequency signals, and each power amplifier module in the second transmission group is connected with the second power module by default; wherein the power supply control circuit is configured to, when the first transmission group has at least one abnormal power amplifier module and the second transmission group does not have an abnormal power amplifier module, switch each normal power amplifier module in the first transmission group from being connected with the currently connected first power module to being connected with the second power module; alternatively, when the second transmission group has at least one abnormal power amplifier module and the first transmission group does not have an abnormal power amplifier module, switch each normal power amplifier module in the second transmission group from being connected with the currently connected second power module to being connected with the first power module.

13. A power supply control method, comprising: acquiring working statuses of individual power amplifier modules in each transmission group of a radio frequency module; and determining, based on the working statuses, an abnormal transmission group in which there is an abnormal power amplifier module, and controlling each normal power amplifier module in the abnormal transmission group to switch from being connected with a currently connected power module to a target power module; wherein the target power module is any one of at least two power modules of the radio frequency module except the currently connected power module; there are at least two transmission groups, each of the transmission groups comprises at least two power amplifier modules, each of the power amplifier modules is switchably connected to any one of the at least two power modules, and the individual power amplifier modules in a same transmission group are configured to perform power amplification on received radio frequency signals under power supply of a same power module.

14. The power supply control method as claimed in claim 13, further comprising: acquiring a target frequency band of a radio frequency signal for which power amplification is supported by the abnormal power amplifier module, and updating an available network search frequency band based on the target frequency band, wherein the available network search frequency band is a frequency band at which the radio frequency module supports for radio frequency signal transmission in a target standard network.

15. The power supply control method as claimed in claim 13, further comprising: disconnecting the abnormal power amplifier module from the currently connected power module.

16. A communication device, comprising a radio frequency module, wherein the radio frequency module comprising: at least two power modules; at least two transmission groups, wherein each of the transmission groups comprises at least two power amplifier modules, each of the power

amplifier modules is switchably connected to any one of the at least two power modules, and the individual power amplifier modules in a same transmission group are configured to perform power amplification on received radio frequency signals under power supply of a same power module; and a power supply control circuit, wherein the power supply control circuit is configured to acquire working statuses of the individual power amplifier modules in each of the transmission groups, determine, based on the working statuses, an abnormal transmission group in which there is an abnormal power amplifier module, and control each normal power amplifier module in the abnormal transmission group to switch from being connected with a currently connected power module to being connected with a target power module, the target power module being any one of the at least two power modules except the currently connected power module.

17. The communication device as claimed in claim 16, wherein the radio frequency module further comprises: a switch module having a controlled terminal, a plurality of first terminals and a plurality of second terminals, wherein each of the plurality of first terminals of the switch module is connected to one of the power modules, at least some of the plurality of first terminals are connected to a same power module, each of the plurality of second terminals of the switch module is connected to a power terminal of one of the power amplifier modules, the controlled terminal of the switch module is connected to the power supply control circuit, and the switch module is configured to switch, under control of the power supply control circuit, the normal power amplifier module in the abnormal transmission group from being connected with the currently connected power module to the target power module; and the switch module is further configured to disconnect, under the control of the power supply control circuit, the abnormal power amplifier module from the currently connected power module.

18. The communication device as claimed in claim 17, wherein the power supply control circuit comprises: a controller, wherein the controller is configured to acquire the working status of each of the power amplifier modules, and generate a power switching instruction in response to the working status being an abnormal working status, the power switching instruction carrying parameter information of the abnormal transmission group and the normal power amplifier module in the abnormal transmission group; and a radio frequency transceiver, wherein the radio frequency transceiver is connected with each of the controller, input terminals of the power amplifier modules, and the controlled terminal of the switch module, and the radio frequency transceiver is configured to provide the radio frequency signals, and control, according to the power switching instruction, the switch module to switch the normal power amplifier module in the abnormal transmission group from being connected with the currently connected power module to being connected with the target power module.

19. The communication device as claimed in claim 18, wherein the radio frequency module further comprises: at least one status detection circuit, wherein the at least one status detection circuit is connected with each of the controller and the power amplifier modules, and is configured to detect the working statuses of the power amplifier modules and feedback the working statuses to the controller.

20. The communication device as claimed in claim 16, wherein the power supply control circuit is further configured to acquire a target frequency band of a radio frequency signal for which power amplification is supported by the abnormal power amplifier module, and update an available network search frequency band based on the target frequency band, the available network search frequency band being a frequency band at which the radio frequency module supports for radio frequency signal transmission in a target standard network.
